

SpatialMosaic: A Multi-View VLM Dataset for Partial Visibility

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<https://kanghee-lee.github.io/spatialmosaic/>

Abstract. The rapid progress of Multimodal Large Language Models (MLLMs) has unlocked the potential for enhanced 3D scene understanding and spatial reasoning. A recent line of work explores learning spatial reasoning directly from multi-view images, enabling MLLMs to understand 3D scenes without explicit 3D reconstructions. Nevertheless, key challenges that frequently arise in real-world environments, such as partial visibility, occlusion, and low-overlap conditions that require spatial reasoning from fragmented visual cues, remain under-explored.

To address these limitations, we propose a scalable multi-view data generation and annotation pipeline that constructs realistic spatial reasoning QAs, resulting in *SpatialMosaic*, a comprehensive instruction-tuning dataset featuring 2M QA pairs. We further introduce *SpatialMosaic-Bench*, a challenging benchmark for evaluating multi-view spatial reasoning under complex and diverse scenarios, consisting of 1M QA pairs across 6 tasks. Our proposed dataset spans both indoor and outdoor scenes, enabling comprehensive evaluation in diverse real-world scenarios. In addition, we introduce a new baseline for multi-view settings, SpatialMosaicVLM, a hybrid framework that integrates 3D reconstruction models as geometry encoders within VLMs for robust spatial reasoning. Extensive experiments demonstrate that our proposed dataset effectively enhances spatial reasoning under challenging multi-view conditions, validating the effectiveness of our data generation pipeline in constructing realistic and challenging QAs. Code and dataset will be available soon.

Keywords: Multi-view Reasoning, Multimodal Large Language Models

1 Introduction

Spatial reasoning in 3D environments is a cornerstone of embodied intelligence, enabling agents to interpret complex scenes and interact effectively with the physical world. The recent progress of MLLMs [2, 19, 22, 46] has raised the possibility of endowing them with human-level 3D spatial understanding, extending their success in 2D perception to more complex tasks such as depth estimation [34, 41, 47], metric distance prediction [8, 34, 41], and holistic spatial reasoning

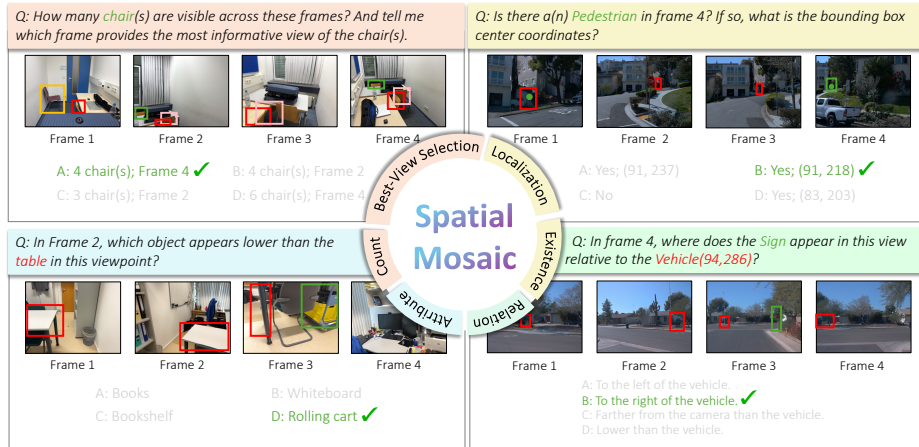


Fig. 1: We present **SpatialMosaic**, a benchmark designed to evaluate 3D spatial reasoning capabilities from fragmented visual cues across multiple viewpoints, spanning both indoor and outdoor scenes. Our benchmark focuses on three challenging real-world scenarios involving partial visibility, occlusion, and low-overlap, where current MLLMs often struggle to maintain geometric and cross-view consistency.

capabilities [11, 34, 41]. However, existing approaches often rely on pre-constructed 3D representations or off-the-shelf reconstruction modules [24, 37]. Such methods require explicit 3D inputs at inference time, which limits their applicability in real-world environments where pre-built 3D maps are unavailable.

To address these limitations, recent studies [34, 41] have investigated deriving 3D spatial reasoning directly from multi-view images, thereby mitigating reliance on pre-constructed geometric priors or conventional 3D reconstruction pipelines. This paradigm not only alleviated such dependencies but also demonstrated superior performance in challenging 3D spatial reasoning tasks. However, existing works still fall short of capturing real-world conditions where available frames are sparse and contain limited overlapping visual information critical for spatial reasoning. In contrast, humans can integrate such partial observations across views to implicitly reconstruct coherent 3D scenes and reason about occluded objects that are not fully visible. Whether MLLMs can achieve comparable robustness under such imperfect conditions remains an open question. Building on these observations, we present fundamentally under-explored scenarios and constraints for 3D spatial reasoning in multi-view systems [25, 36, 44].

In this paper, we define three types of scenarios that each represent a unique spatial reasoning constraint. *Partial visibility* denotes a condition in multi-view settings where an object is visible only in a subset of views, instead of being observed from all camera viewpoints. *Occlusion* refers to a condition within a single view where the object is partially obscured by other instances or truncated by the camera’s field of view. Lastly, *low-overlap* condition represents scenarios

where the available views exhibit minimal cross-view overlap, providing limited information for spatial inference.

To handle these scenarios, we tackle this issue from both data and model perspectives. First, we propose a novel multi-view spatial data generation and annotation pipeline tailored to partial-visibility, occlusion, and low-overlap scenarios. With this pipeline, we construct *SpatialMosaic*, a comprehensive multi-view instruction-tuning dataset containing 2M QA pairs that capture challenging, frequently occurring real-world scenarios. Unlike prior multi-view spatial datasets which focus exclusively on either indoor or outdoor layouts, our dataset spans both domains, enabling more comprehensive evaluation and training across diverse real-world scenes.

While many open-source VLMs can process multi-view inputs, they are not explicitly designed to handle cross-view consistency. In contrast, video-based models are robust to multi-view reasoning but rely on temporally sequential inputs, limiting their practicality in multi-view applications that involve sparse and non-sequential observations. To bridge this gap, we introduce SpatialMosaicVLM, a practical baseline for multi-view settings that integrates 3D reconstruction models as geometry encoders [17, 31–33] within a VLM framework. Finally, we release *SpatialMosaic-Bench*, which provides a more comprehensive and challenging evaluation of spatial capabilities compared to existing multi-view benchmarks [11, 20, 39, 41]. It consists of 1M QA pairs across 6 tasks, focusing on spatial reasoning under realistic and challenging scenarios spanning both indoor and outdoor environments.

The key contributions are summarized as follows:

- We define three under-explored spatial reasoning constraints in multi-view settings, namely *partial visibility*, *occlusion*, and *low-overlap*. Based on these constraints, we propose a scalable data generation pipeline that constructs *SpatialMosaic*, a realistic and challenging instruction-tuning dataset.
- To enable comprehensive evaluation under challenging multi-view conditions, we construct *SpatialMosaic-Bench*, which spans diverse indoor and outdoor environments. By covering both domains, the benchmark facilitates more comprehensive assessment across heterogeneous scene layouts.
- We introduce a new practical baseline for multi-view settings, SpatialMosaicVLM, combining 3D reconstruction models with VLMs to enable cross-view alignment and robust reasoning in realistic, multi-view environments.

2 Related Work

Spatial reasoning with MLLMs. Multimodal large language models (MLLMs) have demonstrated strong capabilities in open-world visual understanding, excelling at classification, segmentation, and captioning. Early models such as Flamingo [2], BLIP-2 [19], MiniGPT-4 [46], and the LLaVA series [22] extend pre-trained large language models (LLMs) with visual encoders, enabling instruction-following and open-ended reasoning on single-image inputs. More recent works seek to move beyond perception by explicitly modeling spatial relations, using

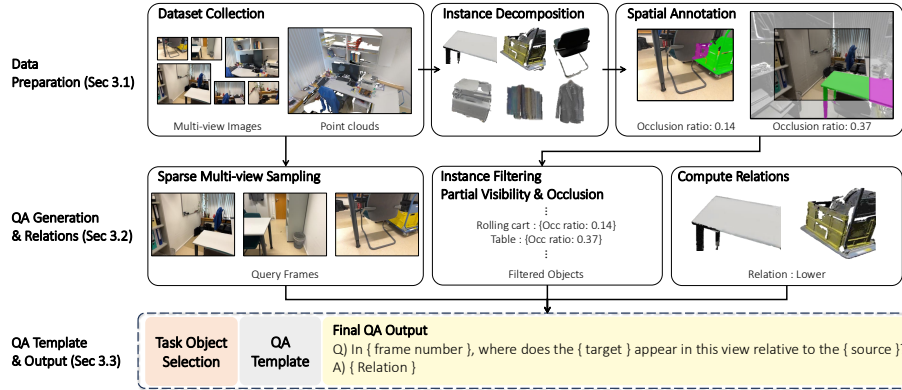


Fig. 2: *SpatialMosaic* data generation pipeline. Given multi-view images and point clouds, we compute various multi-view spatial annotations, including object-level and image-level occlusion ratios for each instance (Sec. 3.1). Images are then filtered by overlap to ensure diverse viewpoints, and instances are filtered based on visibility constraints (Sec. 3.2). Finally, spatial relations are computed and used to populate task-specific templates, generating geometrically grounded QA pairs (Sec. 3.3).

datasets like NLVR2 [40], Neural Module Networks [3], and step-wise reasoning systems including ViperGPT [30] and Text2Scene [16]. Despite these advances, current MLLMs primarily focus on visible 2D cues, limiting their ability to reason when objects are partially visible or occluded in complex spatial configurations.

Towards 3D-aware Vision-Language Models. While 2D-based MLLMs have made progress in spatial reasoning, their reliance on single-view inputs limits the ability to capture full 3D scene structure. Recent approaches incorporate explicit 3D signals, extending vision-language models with depth maps, point clouds, or multi-view consistency. Models such as 3D-LLaVA [9], 3D-LLM [14], Grounded 3D-LLM [5], Scene-LLM [10], LSceneLLM [45] integrate 3D representations for tasks including visual question answering, grounding, and embodied navigation. Transformer-based methods like multi-view transformers for 3D grounding [15] align textual queries with 3D object locations, while benchmarks such as ScanQA [4], SQA3D [23], and ReferIt3D [1] measure performance in real-world settings. Nevertheless, current models remain challenged by cross-view inconsistencies, which degrade reasoning robustness in complex environments.

3D Reconstruction Models. Traditional 3D reconstruction approaches, such as structure-from-motion [7, 13, 26, 27], multi-view stereo methods [12, 28], rely on sequential stages with feature extraction and matching, which provide time-consuming optimization. Recently, large-scale 3D reconstruction models learn generalizable geometric priors from massive multi-view data. Thanks to the rapid advancement of transformers, models such as DUST3R [33], MAST3R [17], CUT3R [32], and VGGT [31] provide 3D point maps with dense correspondences. These models operate over image patches or point tokens, coupled with differentiable geometric modules for epipolar reasoning, multi-view aggregation,

or bundle-adjustment-style optimization, enabling robust generalization across scenes and domains. As a result, these models provide geometry-aware features that transfer across datasets and tasks. Building on this foundation, we integrate model patch tokens with 3D priors to strengthen multi-modal understanding in real-world spatial tasks. This integration improves cross-view consistency, stabilizes object identity under varying appearances, and enables robust reasoning under occlusion. We employ VGGT [31] as a geometry encoder to ground the language model with spatial information by fusing CLIP encoder.

3 SpatialMosaic

While recent VLM benchmarks [34, 35, 41] provide spatial reasoning tasks over multiple images, they either rely on video inputs with sequential frames or do not explicitly address challenges arising from partial visibility, occlusion, and varying image overlap conditions that are pervasive in real-world environments. In practice, these are precisely the scenarios where current VLMs fail, struggling to integrate partial observations and low-overlap images into coherent spatial reasoning. Motivated by these limitations, we first leverage the existing indoor dataset, ScanNet++ [38], to construct challenging spatial reasoning benchmarks. To avoid restricting the pipeline to indoor layouts and to enhance robustness in outdoor layouts, we further extend our dataset to outdoor scenes, Waymo Open Dataset (WOD) [29], thereby encouraging generalization beyond indoor-specific scene structures. We propose a scalable data generation framework that produces more than 3M QA pairs explicitly tailored to partial-visibility, occlusion, and low-overlap scenarios. Unlike prior benchmarks, our dataset includes a larger number of images with substantial perspective changes rather than sequential frames, yielding QA pairs that correspond to more realistic and challenging spatial reasoning tasks commonly encountered in real-world environments. Our data generation pipeline consists of three main stages, as illustrated in Fig. 2: (1) Data Preparation, (2) QA Generation and Relations, and (3) QA Template and Output.

3.1 Data Preparation

Object occlusion ratio. We introduce the object occlusion ratio to quantify the degree of occlusion for each instance. For each scene, ScanNet++ [38] and WOD [29] provide annotated 3D point clouds $\mathcal{P} = \bigcup_{n=1}^N \mathcal{P}_n$, where $\mathcal{P}_n$ represents the point cloud for instance n . We render the complete scene depth map $\mathbf{D}$ and per-instance depth maps $\mathbf{D}_n$ by projecting the provided 3D point cloud onto each camera view using the corresponding camera parameters. Occluded and visible points for instance n are identified by comparing depth values at their projected locations. A point $\mathbf{p}_n \in \mathcal{P}_n$ is occluded when another object blocks the view ($\mathbf{D} < \mathbf{D}_n$), and visible otherwise:

$$\begin{aligned}\mathcal{O}_n &= \{\mathbf{p}_n \in \mathcal{P}_n \mid 0 < \mathbf{D} < \mathbf{D}_n\} \\ \mathcal{V}_n &= \{\mathbf{p}_n \in \mathcal{P}_n \mid \mathbf{D}_n \leq \mathbf{D}, \mathbf{D}_n < \infty\}.\end{aligned}\tag{1}$$

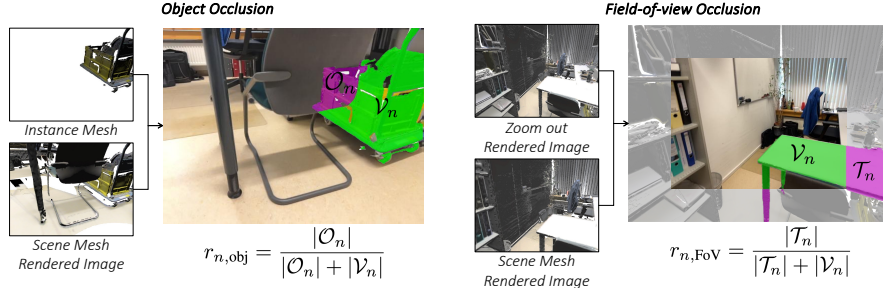


Fig. 3: Occlusion ratio calculation. We render each instance independently to measure visible (green) and occluded (magenta) pixels. **Object Occlusion:** Object-level occlusion (r_{obj}) captures inter-object obstruction from the actual camera view. **Field-of-view Occlusion:** Field-of-view truncation (r_{FoV}) uses extended field-of-view rendering to quantify boundary occlusion from frame cropping.

The object occlusion ratio is defined as:

$$r_{n,\text{obj}} = \frac{|\mathcal{O}_n|}{|\mathcal{O}_n| + |\mathcal{V}_n|} \quad (2)$$

where the object occlusion ratio $r_{n,\text{obj}} \in [0, 1]$ represents the proportion of occluded points for instance n , as illustrated in Fig. 3.

FoV occlusion ratio. In addition to object-level occlusion, instances may be partially occluded due to field of view (FoV) truncation, where parts of the object extend beyond the camera’s field of view. To quantify this, we introduce the FoV occlusion ratio, which measures the proportion of the instance truncated by the image boundary. As illustrated in Fig. 3, we create an extended field of view reference image with doubled resolution $2H \times 2W$ centered around the original view by shifting the principal point to the center of the extended field. The extended intrinsic matrix $\tilde{\mathbf{K}} \in \mathbb{R}^{3 \times 3}$ is defined as $\tilde{\mathbf{K}} = [f_x, 0, c_x + W/2; 0, f_y, c_y + H/2; 0, 0, 1]$ where f_x, f_y are the focal lengths and c_x, c_y are the principal point coordinates from the original intrinsic matrix $\mathbf{K}$. Using this extended intrinsic matrix, we project each point to obtain $(\tilde{u}, \tilde{v})$ in the extended image coordinate system. We then render both the instance depth map $\tilde{\mathbf{D}}_n$ and the complete scene depth map $\tilde{\mathbf{D}}$ in this extended view.

We define the visible point sets within the original FoV region $\mathcal{R}$ and the FoV-truncated region $\mathcal{T}_n$ as follows:

$$\begin{aligned} \mathcal{V}_n &= \{\mathbf{p}_n \in \mathcal{P}_n \mid 0 < \tilde{\mathbf{D}}_n \leq \tilde{\mathbf{D}} < \infty, (\tilde{u}, \tilde{v}) \in \mathcal{R}\} \\ \mathcal{T}_n &= \{\mathbf{p}_n \in \mathcal{P}_n \mid 0 < \tilde{\mathbf{D}}_n \leq \tilde{\mathbf{D}} < \infty, (\tilde{u}, \tilde{v}) \in \tilde{\mathcal{R}} \setminus \mathcal{R}\} \end{aligned} \quad (3)$$

where $\mathcal{R} = [W/2, 3W/2) \times [H/2, 3H/2)$ represents the original FoV region and $\tilde{\mathcal{R}} = [0, 2W) \times [0, 2H)$ represents the extended canvas region. The FoV occlusion ratio is then defined as:

$$r_{n,\text{FoV}} = \frac{|\mathcal{T}_n|}{|\mathcal{T}_n| + |\mathcal{V}_n|} \quad (4)$$

where $r_{n,\text{FoV}} \in [0, 1]$ quantifies the proportion of points truncated by the FoV boundaries, with $r_{n,\text{FoV}} = 0$ indicating the instance is fully within the original field of view.

3.2 QA Generation and Relations

Sparse Multi-view Sampling. To construct multi-frame spatial reasoning tasks, we sample image pairs with limited overlap to encourage integration of diverse viewpoints. For each scene, we compute the overlap ratio between i, j th images by measuring the intersection-over-union of their visible 3D points. The overlap ratio is defined as follows,

$$\text{Overlap}(i, j) = \frac{|\mathcal{V}^i \cap \mathcal{V}^j|}{|\mathcal{V}^i \cup \mathcal{V}^j|}, \quad \mathcal{V} = \bigcup_{n=1}^N \mathcal{V}_n. \quad (5)$$

We retain only image pairs with overlap ratios below τ , thereby filtering out redundant images with excessive shared content. This constraint encourages integrative spatial reasoning across sparse and diverse viewpoints rather than simple matching of overlapping regions.

Instance Filtering. Not all instances are suitable for generating meaningful QA pairs. We filter instances based on two criteria: (1) the target instance must not appear in all images within the selected combination to ensure partial visibility, and (2) the target instance must have an occlusion ratio below 0.9 in at least one image (objects with occlusion ratios above 0.9 are nearly impossible for humans to observe). Additionally, there are task-specific filtering criteria such as selecting objects as target instances only if it is above the minimum occlusion threshold (e.g., above 0.4), and ensuring that the source instance is visible while the target instance is not visible in the selected query image. Collectively, these constraints substantially elevate the spatial complexity of the generated QAs; they reduce reliance on redundant visibility cues, force models to reason under minimal information with asymmetric view conditions, and require accurate spatial inference even when object instances exhibit no correlation under a single image. As a result, our QA set captures challenging real-world visibility patterns that conventional datasets fail to represent.

Compute Relations. To compute relations between objects, we extract each object’s 3D oriented bounding box and integrate it with camera coordinates to compute positional relationships along the X, Y, and Z axes in the image. These relationships define the underlying reasoning context, allowing us to determine directional relations (e.g., "to the left side, above") between object pairs. The computed relations serve as ground-truth labels for spatial reasoning questions.

3.3 QA Template and Output

Task and Object Selection. Based on the filtered instances and computed relations, we select appropriate source and target objects for each task type. The

selection process considers task requirements and ensures that selected objects satisfy visibility and occlusion constraints established in earlier stages.

QA Template Generation. We define templates for 6 task categories using template-based generation, where placeholders such as [source_obj], [target_obj], or [relation] are replaced with concrete object labels and computed spatial information. The tasks include: Object Count, Best-View Selection, Object Localization, Occlusion-Aware Object Existence, Occlusion-Aware Attribute, and Occlusion-Aware Spatial Relation. Fig. 1 illustrates representative examples for each task category. Answer options are generated automatically according to the predefined QA format: binary questions contain one correct and one inverted relation, while four-option multiple-choice questions include one correct answer and three distractor relations with incorrect spatial configurations. The ground-truth label is calculated in accordance with the relation or image, consistent with the 3D metadata and visibility constraints. For counting-based tasks, we track instance IDs across frames to ensure identical objects from different viewpoints are counted only once. Additional QA templates and detailed generation pipelines are provided in the Appendix.

3.4 Bias Analysis

We analyze potential biases that may arise during the dataset construction process. To mitigate such biases, we carefully design the benchmark and conduct empirical analyses. (1) To prevent answer-option imbalance in Multiple-Choice Question(MCQ) tasks, the position of the correct answer is randomly assigned while enforcing a uniform distribution over all answer choices. This prevents the model from relying on trivial answer-selection heuristics, such as favoring a specific option index, and ensures that performance reflects the model’s reasoning capability rather than positional biases. (2) Unlike prior VQA datasets that predominantly generate QA pairs from frequently occurring object categories, we construct a uniform number of QA pairs across all available categories. By leveraging the full set of object categories rather than focusing on common instances, we increase semantic diversity and mitigate skewed label distributions and object co-occurrence biases. This design encourages models to leverage geometric information for prediction rather than relying on category priors or language cues. (3) Query frames are randomly sampled from candidate frames, while answers are determined by explicit geometric criteria. For example, in best-view selection tasks, the correct answer is derived from geometric signals such as visible object instances and pixel coverage to reduce frame-index and viewpoint biases. (4) Finally, we examine whether performance is influenced by the aforementioned biases or dataset priors rather than reasoning capability. We observe that model performance consistently decreases as task difficulty increases, requiring more complex cross-view geometric reasoning (see Appendix for details). These results indicate that accuracy is driven by the model’s multi-view reasoning capability rather than by bias-induced shortcuts or dataset priors.

3.5 SpatialMosaic Benchmark

SpatialMosaic-Bench is a large-scale multi-view benchmark explicitly designed to evaluate spatial reasoning under occlusion and partial visibility. The benchmark comprises 1M QA pairs across 6 task categories derived from real-world indoor and outdoor scenes in ScanNet++ [38] and WOD [29].

Evaluation protocol. Each QA instance consists of 2-5 frames, a question, and multiple answer options. Models select one option, and performance is measured by accuracy against ground-truth answers. To enable fine-grained analysis, we annotate every QA with two diagnostic scenarios: (1) *Visibility Scenario* indicates whether target objects are consistently visible or partially occluded across frames, and (2) *GT Scenario* indicates whether all instances of the target category in the scene are captured or only a subset is visible. Together, these axes support structured performance breakdowns across a continuum of difficulty levels. In particular, the combination *Partially Visible + Partial Coverage* corresponds to the most challenging regime, where models must aggregate sparse cues, resolve viewpoint inconsistencies, and reason over incomplete category-level information. This diagnostic framework enables us to analyze not only overall accuracy but also the specific multi-view conditions under which models succeed or fail. Additional evaluation details are provided in the Appendix.

4 SpatialMosaicVLM Architecture

We integrate VGGT [31], a 3D reconstruction model, into our VLM architecture as a spatial encoder. VGGT [31] provides geometry-aware token representations that are fused with per-view visual tokens from the image encoder. This joint representation captures multi-view consistency while maintaining geometric structure, thereby supporting spatial reasoning across viewpoints.

To leverage this joint representation, we extract and fuse visual and geometric tokens from both the encoder and decoder stages as illustrated in Fig. 4. Specifically, each image $\mathbf{I}$ passes through both the visual encoder E_{vis} and geometric encoder E_{geo} to obtain visual tokens F_{vis} and geometric tokens F_{geo} , respectively:

$$F_{vis} = E_{vis}(\mathbf{I}), \quad F_{geo} = E_{geo}(\mathbf{I}). \quad (6)$$

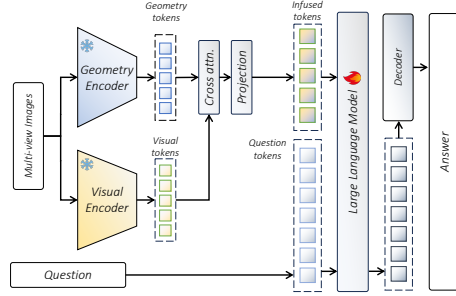


Fig. 4: SpatialMosaicVLM architecture. Multi-view images are processed by parallel Geometry and Visual Encoders to extract 3D structural and appearance features. The resulting geometry and visual tokens are fused via cross-attention, then passed to the language model with question tokens to answer spatial reasoning questions under occlusion and partial visibility circumstances.

We then fuse the visual tokens F_{vis} with geometric tokens F_{geo} via cross-attention to obtain 3D aware visual tokens :

$$F_{fuse} = \sigma \left(\frac{(F_{vis}W_q)(F_{geo}W_k)^T}{\sqrt{d_k}} \right) (F_{geo}W_v) \quad (7)$$

where W_q, W_k, W_v are learnable projection matrices, d_k is the key dimension and σ is softmax operation. Following LLaVA-Next-Video [43], the fused tokens F_{fuse} pass through a two-layer projector to obtain F'_{fuse} , which are concatenated with question tokens $F_{question}$ as input to the language model backbone.

5 Experiments

Datasets. We evaluate our approach across two complementary spatial benchmarks, each examining a distinct aspect of multi-view spatial reasoning. First, we analyze model performance on *SpatialMosaic-Bench*, which consists of two evaluation splits: *SpatialMosaic-Indoor* (Sec. 5.1) and *SpatialMosaic-Outdoor* (Sec. 5.3). *SpatialMosaic-Indoor* is constructed from ScanNet++ [38], while *SpatialMosaic-Outdoor* is built upon the Waymo dataset [29]. These benchmarks assess robustness in realistic environments spanning both indoor and outdoor scenes, involving partial visibility, occlusion, and low-overlap conditions. *SpatialMosaic-Bench* offers a challenging evaluation protocol by constructing view sets with deliberately minimal geometric redundancy. It is designed to probe robustness under incomplete spatial evidence and to reflect more realistic low-overlap conditions. Next, we further evaluate our model on VSI-Bench [35], an informative benchmark designed to assess conventional spatial understanding capabilities. It includes spatial configuration tasks, such as object counting, relative distance and direction estimation, as well as measurement-related tasks including room size, object size, and absolute distance prediction.

Baselines. Following prior efforts in spatial reasoning, we evaluate our method against a suite of open-sourced Video-Language Models, which serve as our baselines under identical multi-view input protocols. The comparing models include LLaVA-OneVision-0.5B [18], InternVL2-2B [6], LLaVA-NeXT-Video-7B [43], LLaVA-OneVision-7B [18], LongVA-7B [42], InternVL2-8B [6], VILA-1.5-8B [21], and VILA-1.5-40B [21]. All baselines receive the same set of multi-view frames at a resolution of 518×518 per view, ensuring consistent evaluation across architectures. Model outputs are derived using their default decoding settings without task-specific tuning.

Implementation Details. For training all ablated models, we employ 8 NVIDIA H200 GPUs with a batch size of 4. We used accelerator library with DeepSpeed ZeRO stage 2 optimization for distributed training. The learning rate is set to 2×10^{-5} , the weight decay to 0.0, and we adopt a cosine learning rate scheduler. Training is performed for 5 epochs. Both the visual and geometry encoders are frozen, and multi-view features are integrated through a 3D-fusion module composed of a cross-attention layer followed by a projection layer.

Table 1: Quantitative results on SpatialMosaic-Indoor. **Bold** and underline indicate the best and second-best performance within open-sourced VLMs for each task, respectively. **Highlighting** denotes the top-3 ranked models overall.

Methods	Rank	Avg.	Obj. Count	Best View.	Obj. Exist.	Obj. Att.	Obj. Rel.	Obj. Loc.
<i>SpatialMosaic-tiny Perf.</i>								
Human	-	55.1	70.0	40.0	66.6	41.1	50.0	63.3
LLaVA-NeXT-Video	-	49.3	70.0	50.0	60.0	24.4	58.3	43.3
<i>Open-sourced VLMs</i>								
LLaVA-OneVision-0.5B	9	37.7	29.5	44.4	55.3	20.7	37.7	38.3
InternVL2-2B	6	39.9	<u>65.8</u>	49.4	47.8	26.4	34.8	<u>53.1</u>
LLaVA-NeXT-Video-7B	3	47.8	61.1	41.0	45.7	34.2	<u>54.6</u>	37.2
LLaVA-OneVision-7B	5	42.8	58.5	37.5	56.1	32.6	37.9	37.8
LongVA-7B	8	38.2	34.5	27.5	57.4	24.2	42.4	26.2
InternVL2-8B	4	46.0	61.6	49.0	54.4	<u>38.6</u>	39.8	43.3
VILA-1.5-8B	7	38.7	40.2	24.7	52.5	32.5	37.5	32.4
VILA-1.5-40B	2	<u>48.5</u>	56.0	<u>50.7</u>	<u>58.9</u>	31.6	54.2	40.6
SpatialMosaicVLM (7B)	1	77.4	89.9	72.9	81.5	74.3	84.0	61.8

5.1 Evaluation on SpatialMosaic-Indoor

SpatialMosaic-Bench provides a challenging and realistic evaluation characterized by partial visibility, occlusion, and minimal overlap across viewpoints. These conditions limit geometric redundancy and require models to infer spatial structure from fragmented observations rather than relying on stable cross-frame correspondences. Table 1 reports the evaluation results on the indoor split of *SpatialMosaic-Bench*. Despite their strong spatial reasoning ability in conventional image or video settings, existing MLLM baselines struggle under these conditions. In particular, models frequently misinterpret object presence, confuse spatial relations, or fail to localize objects across sparsely aligned viewpoints. These results highlight the difficulty of integrating fragmented visual evidence across views when spatial cues are incomplete or partially observed. Additionally, we evaluate the most challenging split using SpatialMosaic-tiny, a reduced version of the full benchmark consisting of 300 randomly selected questions, to benchmark against human performance. Fine-tuning of SpatialMosaicVLM is conducted using the training split of *SpatialMosaic*, while all models are evaluated on *SpatialMosaic-Bench* under the same evaluation protocol and data split.

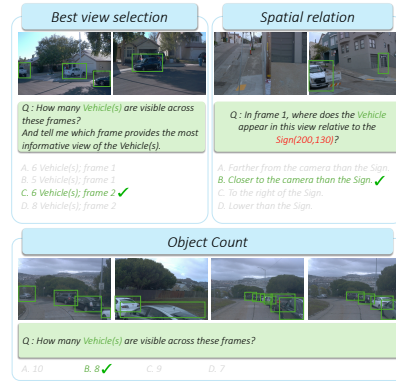


Fig. 5: SpatialMosaic-Outdoor.

Table 2: Quantitative results on VSI-Bench. Comparison with strong baselines across diverse spatial reasoning tasks. SpatialMosaicVLM achieves the best overall performance, outperforming 72B-scale models while using significantly fewer parameters.

Methods	Rank	Avg.	Obj. Count	Abs. Dist.	Obj. Size	Room Size	Rel. Dist.	Rel. Dir.	Route Plan	Appr. Order
			Numerical Answer			Multiple-Choice Answer				
Proprietary Models (API)										
GPT-4o	-	34.0	46.2	5.3	43.8	38.2	37.0	41.3	31.5	28.5
Gemini-1.5 Flash	-	42.1	49.8	30.8	53.5	54.4	37.7	41.0	31.5	37.8
Gemini-1.5 Pro	-	45.4	56.2	30.9	64.1	43.6	51.3	46.3	36.0	34.6
Open-sourced VLMs										
LLaVA-OneVision-0.5B	11	28.0	46.1	28.4	15.4	28.3	28.9	36.9	34.5	5.8
InternVL2-2B	12	27.4	21.8	24.9	22.0	35.0	33.8	44.2	30.5	7.1
LLaVA-NeXT-Video-7B	5	35.6	48.5	14.0	47.8	24.2	43.5	42.4	34.0	30.6
InternVL2-8B	6	34.6	23.1	28.7	48.2	39.8	36.7	30.7	29.9	39.6
LLaVA-OneVision-7B	7	32.4	47.7	20.2	47.4	12.3	42.5	35.2	29.4	24.4
LongVA-7B	9	29.2	38.0	16.6	38.9	22.2	33.1	43.3	25.4	15.7
VILA-1.5-8B	10	28.9	17.4	21.8	50.3	18.8	32.1	34.8	31.0	24.8
LongVILA-8B	13	21.6	29.1	9.1	16.7	0.0	29.6	30.7	32.5	25.5
InternVL2-40B	4	36.0	34.9	26.9	46.5	31.8	42.1	32.2	34.0	39.6
VILA-1.5-40B	8	31.2	22.4	24.8	48.7	22.7	40.5	25.7	31.5	32.9
LLaVA-NeXT-Video-72B	2	40.9	48.9	22.8	57.4	35.3	42.4	36.7	35.0	48.6
LLaVA-OneVision-72B	3	40.2	43.5	23.9	57.6	37.5	42.5	39.9	32.5	44.6
SpatialMosaicVLM (7B)	1	59.6	70.6	48.6	70.1	65.2	60.4	77.5	43.8	40.5

5.2 Evaluation on VSI-Bench

To further examine conventional spatial reasoning capability beyond the task formulations defined in *SpatialMosaic-Bench*, we evaluate our model on VSI-Bench. Unlike our benchmarks, which emphasize multi-view reasoning under challenging conditions, VSI-Bench primarily focuses on standard spatial reasoning tasks, such as relative distance estimation, directional reasoning, and object size comparison. As shown in Tab. 2, SpatialMosaicVLM achieves the best overall performance among all evaluated models, substantially outperforming strong proprietary and open-source VLM baselines. Notably, SpatialMosaicVLM outperforms 72B-scale model [43] by 18.7%p, despite using nearly $10\times$ fewer parameters. The performance gains are consistent across diverse task categories, indicating that SpatialMosaicVLM maintains strong multi-view reasoning capability across various spatial reasoning tasks. Although VSI-Bench does not explicitly focus on occlusion-heavy or low-overlap scenarios, SpatialMosaicVLM maintains strong performance across its task categories, demonstrating strong spatial reasoning capability across both challenging multi-view settings and more conventional benchmarks.

5.3 Evaluation on SpatialMosaic-Outdoor

Existing VLM benchmarks are often confined to a single environmental domain, typically focusing exclusively on either indoor layouts or outdoor scenes. Such

Table 3: Quantitative results on SpatialMosaic-Outdoor. Zero-shot evaluation on outdoor scenes constructed from the Waymo dataset.

Methods	Rank	Avg.	Obj. Count	Best View.	Obj. Exist.	Obj. Rel.	Obj. Loc.
<i>Open-sourced VLMs</i>							
LLaVA-NeXT-Video-7B	3	44.2	54.7	40.2	48.5	53.5	37.5
LLaVA-OneVision-7B	6	35.9	26.4	22.6	57.3	35.4	31.8
LongVA-7B	5	36.0	38.3	23.6	44.2	45.5	29.2
InternVL2-8B	4	42.4	45.3	43.6	49.7	49.1	39.0
VILA-1.5-8B	7	33.8	40.8	22.9	50.6	29.5	37.0
VILA-1.5-40B	2	47.9	41.8	43.7	60	59.0	37.0
SpatialMosaicVLM (7B)	1	62.0	80.0	67.0	73.0	65.3	58.4

domain-specific evaluation settings limit the ability to assess whether a model has acquired transferable multi-view spatial reasoning capabilities, as performance may be influenced by recurring structural patterns inherent to a particular layout distribution. To address this limitation, we extend our evaluation to encompass both indoor and outdoor environments. In particular, we evaluate *SpatialMosaic-Outdoor* in a zero-shot setting, without fine-tuning on outdoor data, to examine out-of-domain generalization. This benchmark is constructed using the large-scale Waymo dataset [29], introducing complex outdoor driving scenarios with substantially different scene geometry and spatial configurations compared to indoor environments. Despite this domain shift, our model maintains strong performance across task categories. These results indicate that the learned spatial representations transfer effectively beyond the indoor domain, demonstrating robust out-of-domain spatial reasoning capability. Overall, the consistent performance across both *SpatialMosaic-Indoor* and *SpatialMosaic-Outdoor* suggests that our approach mitigates layout-specific bias and supports reliable zero-shot generalization across heterogeneous environments.

5.4 Ablation Studies

We conduct ablation studies to evaluate the efficiency of our model, which serves as a new baseline for the multi-view setting. For a fair comparison, all ablated models are fine-tuned using the same training data and follow identical optimization and decoding settings. Evaluations are conducted on *SpatialMosaic-Bench*, and VSI-Bench under the same protocol as our SpatialMosaicVLM model. To examine the contribution of the proposed geometric encoder, we compare the SpatialMosaicVLM with a variant that removes the geometry encoder. As shown in Tables 4 and 5, removing geometric features consistently degrades performance across both *SpatialMosaic-Bench* and VSI-Bench. The performance drop is particularly noticeable in tasks that require explicit spatial reasoning, such as object attribute understanding, spatial relations, and directional reasoning. These tasks depend heavily on geometric cues for resolving spatial configuration across

Table 4: Ablation study on *SpatialMosaic-Bench*.

Methods	Avg.	Obj. Count.	Best View	Obj. Exist.	Obj. Attr.	Obj. Rel.	Obj. Loc.
<i>SpatialMosaic-Bench.</i>							
LLaVA-NeXT-Video (w/o Geo. enc.)	76.5	89.5	72.5	80.6	72.4	82.7	61.2
SpatialMosaicVLM	77.4	89.9	72.9	81.5	74.3	84.0	61.8

Table 5: Ablation study on VSI-Bench.

		<i>Obj. Count</i>	<i>Abs. Dist.</i>	<i>Obj. Size</i>	<i>Room Size</i>	<i>Rel. Dist.</i>	<i>Rel. Dir.</i>	<i>Route Plan</i>	<i>Appr. Order</i>
Methods	Avg.	Numerical Answer				Multiple-Choice Answer			
<i>VSI-Bench.</i>									
LLaVA-NeXT-Video (w/o Geo. enc.)	54.8	69.7	44.4	70.0	60.3	57.3	55.4	39.2	42.1
SpatialMosaicVLM	59.6	70.6	48.6	70.1	65.2	60.4	77.5	43.8	40.5

views. Without the geometry encoder, the model struggles to accurately infer object properties and spatial relationships from multi-view observations. Overall, these results demonstrate that incorporating explicit geometric representations significantly improves the model’s ability to reason about spatial structure, highlighting the importance of geometric cues for robust multi-view spatial understanding.

6 Conclusion

In this work, we take an initial step toward tackling the challenges of partial visibility, occlusion, and low-overlap conditions in settings that require models to integrate fragmented visual cues to form coherent 3D understanding. To support this goal, we introduced an automatic multi-view data generation pipeline, enabling the construction of *SpatialMosaic* and *SpatialMosaic-Bench*, which capture challenging multi-view scenarios across indoor and outdoor scenes. We further introduce a practical baseline for multi-view settings, SpatialMosaicVLM, a hybrid framework that combines geometric cues from 3D reconstruction models to enable effective cross-view alignment and robust spatial reasoning. Experiments demonstrate that instruction-tuning on our dataset improves performance under incomplete visual evidence. These results highlight the importance of equipping models with the ability to aggregate partial observations and infer coherent 3D structure from limited cues. We believe that this work enhances the scalability and real-world applicability of MLLMs, contributing to narrowing the gap toward human-level multi-view reasoning.

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A Additional Experiments

To further validate the efficiency of our design beyond indoor environments, we conduct an additional ablation study on SpatialMosaic-Outdoor, which extends our benchmark to large-scale outdoor scenes. Following the same protocol as in Sec. 5.4, all ablated models are fine-tuned using identical training data, optimization, and decoding settings to ensure a fair comparison. Table 6 shows that SpatialMosaicVLM generally outperforms its geometry-removed variant on SpatialMosaic-Outdoor, achieving higher average performance and gains on most tasks. The gains are particularly notable in tasks involving relational reasoning, such as Existence and Spatial Relation. This pattern is consistent with the indoor ablations in Tables 4 and 5, further confirming that the proposed geometry encoder contributes to robust multi-view spatial reasoning across both indoor and outdoor layouts.

Table 6: Ablation study on SpatialMosaic-Outdoor.

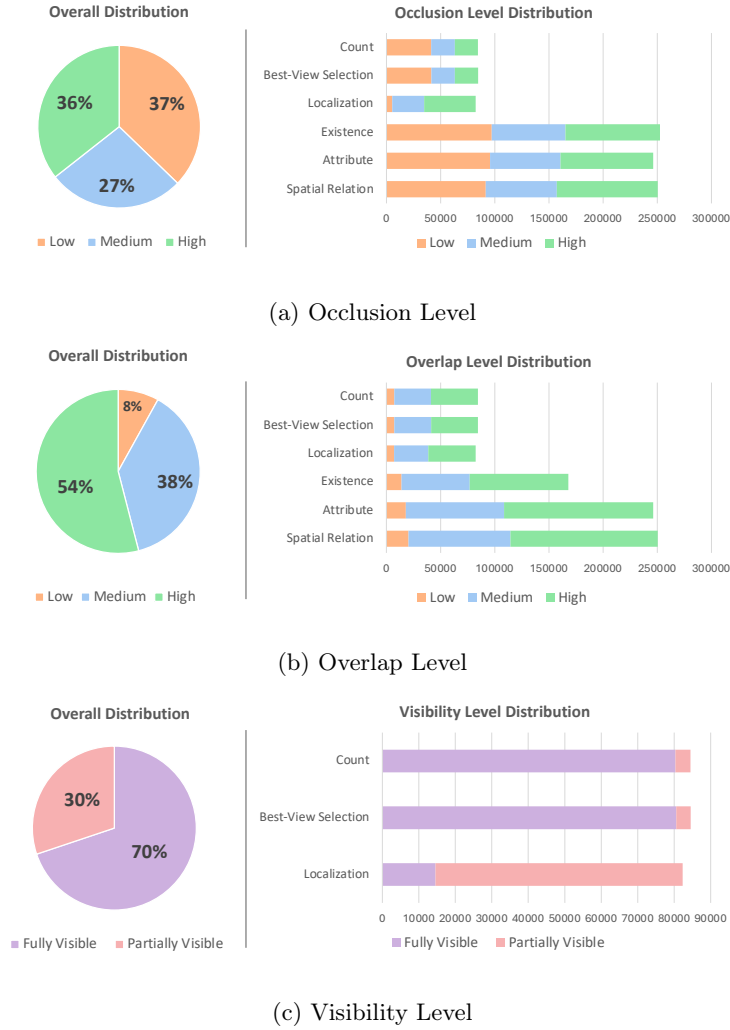
Methods	Avg.	<i>Obj. Count.</i>	<i>Best View</i>	<i>Obj. Exist.</i>	<i>Obj. Rel.</i>	<i>Obj. Loc.</i>
<i>SpatialMosaic-Outdoor.</i>						
LLaVA-NeXT-Video (w/o Geo. enc.)	66.65	60.17	69.14	67.24	67.57	66.09
SpatialMosaicVLM	70.34	64.66	69.31	72.79	71.13	64.61

B Statistics of SpatialMosaic-Bench

We provide detailed statistics of *SpatialMosaic-Bench* across different difficulty levels in Fig. 6. Our benchmark, spanning both indoor and outdoor scenes, contains a total of 1M QA pairs distributed across six main task categories: Count, Best-View Selection, Existence, Attribute, and Localization. To ensure a comprehensive evaluation of spatial reasoning capabilities, we emphasize challenging scenarios by generating more samples from high and medium difficulty levels. Note that we enforce the target object to be invisible in the query frame for attribute, existence, and spatial relation tasks; thus, all QA samples from these tasks fall under the Partially Visible category in the visibility-level distribution.

C Analysis under Different Conditions

We evaluate SpatialMosaicVLM on *SpatialMosaic-Bench* under varying difficulty levels: Occlusion, Overlap, and Visibility. Performance consistently declines as the difficulty increases, confirming that our categorization accurately captures the challenges of multi-view spatial reasoning. We further compare SpatialMosaicVLM with InternVL2-8B [6] across all tasks (Fig. 7). **Visibility Level.** For the Object count task, performance drops significantly on the Partially Visible condition, reflecting the inherent difficulty of counting objects that are not fully visible in

**Fig. 6:** Difficulty Level Distribution

every frame. For Best-View Selection combined with counting (Bestview count), both Partially and Fully Visible cases show similar moderate performance, as the model must both count instances and identify the optimal frame. Our model substantially outperforms InternVL2-8B on Partially Visible cases across all tasks. **Overlap & Occlusion Level.** For Attribute, Existence, and Spatial Relation tasks, performance consistently declines as task difficulty increases (from Low to High levels), with InternVL2-8B exhibiting a larger performance drop. In contrast, Count and Best-View Selection tasks are not significantly affected by low overlap and occlusion conditions. Instead, we empirically observe that their performance

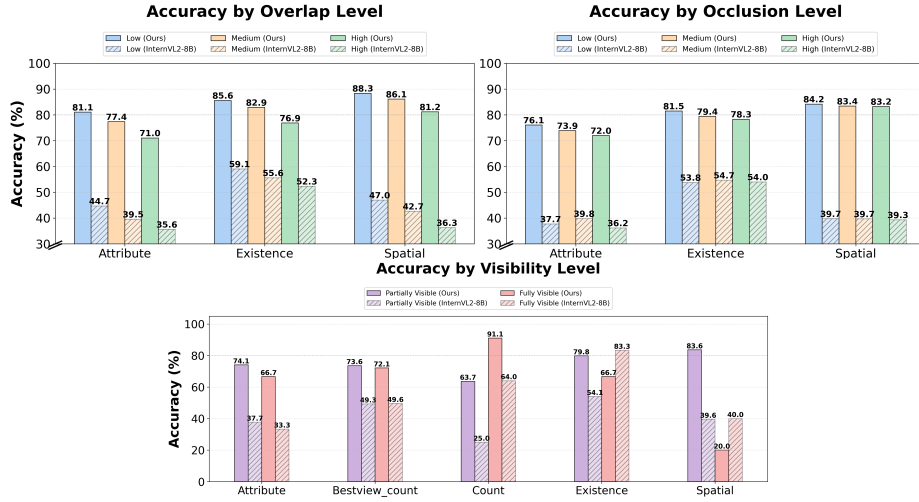


Fig. 7: Comparison between SpatialMosaicVLM and InternVL2-8B on *SpatialMosaic-Bench*.

is primarily influenced by two factors: the total number of visible instances of the target object category across all frames and the frame-wise distribution of visible instance counts.

D Technical Details for SpatialMosaic

D.1 Spatial Annotation Framework

The spatial annotation framework establishes the geometric signals used throughout the VQA generation pipeline in *SpatialMosaic-Bench*. In this section, we provide a technical overview of the annotation process introduced in Sec. 3, focusing on how instance-level visibility, occlusion, and multi-view overlap are computed and stored for later use. We directly reference the definitions in Sec. 3 and clarify how these quantities are applied in the downstream QA-generation stages.

Instance visibility. For each object instance, visibility is determined using the object-level occlusion ratio and FoV occlusion ratio defined in Eq. (1) - (4) of Sec. 3.1. Using the ScanNet++ point clouds and calibrated camera parameters, each instance’s 3D points are projected into every view, and visibility is computed by rendering both the scene-level depth map and per-instance depth map as described in Eq. (1). FoV truncation is computed via the extended intrinsic formulation in Eq. (3) - (4). These two metrics together form the per-frame visibility profile used in all subsequent stages.

Per-frame visibility masks. Using the visible point sets defined in Eq. (1) and Eq. (3), we construct binary masks for each instance in each frame. These masks

define which portion of the 3D geometry is observable and directly ground the multi-view filtering and relation computation.

Overlap computation. Frame overlap is computed using the intersection-over-union of visible 3D point sets as introduced in Eq. (5). This ensures that multi-view sampling is guided by geometric overlap rather than superficial image similarity. Following Sec. 3.2, we construct VQA samples using only image pairs whose overlap ratio is below a predefined threshold τ . To determine an appropriate value of τ , we conduct both quantitative and qualitative analyses on a subset of the data. Figure 9 provides a quantitative analysis of model performance across different overlap ratios. Accuracy varies substantially within the 10–30% overlap range, indicating meaningful differences in task difficulty. Based on this observation, we set $\tau = 0.3$ to retain sufficiently challenging low-overlap pairs while avoiding overly easy cases. Figure 8 presents qualitative examples of image pairs with varying overlap ratios.

Bounding box transformation. As described in Sec. 3.2, spatial relations are computed by comparing the positions of objects in the viewpoint of the selected query frame. Since the relation is determined by evaluating how the oriented bounding boxes of two instances are ordered along the camera-frame axes, each bounding-box vertex is first transformed into the camera coordinate system of the query view. The transformation is: $v^{(c)} = R_{wc}(v - t_{wc})$ where v is a bounding-box vertex in world coordinates, R_{wc} is the world-to-camera rotation matrix, and t_{wc} is the camera-center translation. The resulting vertices $v^{(c)}$ define the camera-frame bounding boxes used for axis-aligned separation when computing the directional relation in necessary tasks.

D.2 Data Generation Pipeline

The data generation pipeline constructs QA samples by applying task-dependent constraints on the annotated geometric information in Sec. D.1. Once visibility, occlusion, and overlap statistics are available, the pipeline proceeds through the following steps:

Frame combination construction. For each scene, candidate multi-view combinations are formed by enumerating frame sets that satisfy the required view count and the overlap constraint. Only combinations whose internal view-overlap stays below the specified threshold are retained, ensuring sparse and complementary viewpoints.

Valid instance set collection. Within each retained combination, all object instances that appear in at least one of the included frames are collected. Instances that violate the partial-visibility requirement, such as appearing in every frame or remaining nearly fully occluded across all frames, are removed. The remaining instances carry their semantic labels, visibility flags, and camera-frame bounding boxes.

Query-frame and object selection. A query frame is randomly selected from the valid combinations. Depending on the task, the pipeline verifies whether the instances in that frame satisfy the required visibility conditions (e.g., visible source and invisible target pairs for multi-category tasks, or category uniqueness

for localization). If the condition is not met, the pipeline attempts another configuration. The task-specific visibility conditions are elaborated in Sec. E.

Task-specific geometric computation. Geometric quantities are computed only after a valid configuration is found. These computations are task-specific, including directional separations between instance bounding boxes, visible-pixel statistics, or merged instance counts across views. All computations operate directly on the pre-annotated camera-frame geometry.

Answer and distractor generation. The computed values are inserted into the task templates. Distractors are generated using the rules defined for each task, such as orthogonal-axis relations or offset-based count alternatives. All options are de-duplicated, validated, and randomized.

QA assembly. The final QA entry records the selected frame combination, the question constructed from the template with appropriate instances, the multiple-choice options, and all geometric metadata required for evaluation. After such a process is completed, a single QA is generated, and the pipeline then moves on to the next sample.

E SpatialMosaic Task Descriptions

Algorithm 1: BaseQAGenerator

```

1 Input: Scene list  $\mathcal{S}$ , scene-level metadata  $\mathcal{M}$ , frame-level metadata  $\mathcal{F}$ 
2 Output: Generated QA set  $\mathcal{Q}$ 
3 Initialization:
4   Build per-scene overlap tables from  $\mathcal{F}$ 
5   Initialize QA buffers and global counters
6 Main loop over scenes:
7 for  $s \in \mathcal{S}$  do
8    $F_s \leftarrow \mathcal{F}[s]$ 
9    $M_s \leftarrow \mathcal{M}[s]$ 
10   $OV_s \leftarrow read\_overlap\_table\mathcal{F}[s]$ 
11  % Step 1. Occlusion extraction:
12   $(O_s, L_s) \leftarrow extract\_occlusion(F_s)$ 
13  % Step 2. Frame-combination construction:
14   $C_s \leftarrow sample\_valid\_combos(O_s, L_s, s)$ 
15   $C_s \leftarrow overlap\_filtering(C_s, OV_s)$ 
16  % Step 3. Scene-level QA generation:
17  for  $t \in T$  do
18     $G_t \leftarrow get\_qa\_generator(t)$ 
19     $Q_s \leftarrow G_t(s, M_s, F_s, C_s)$ 
20  end
21  % Step 4. Accumulate results:
22   $\mathcal{Q} \leftarrow \mathcal{Q} \cup Q_s$ 
23 end

```

The BaseQAGenerator serves as a foundation for the QA generation process. For each frame in the scene, instance-level occlusion ratios O_s , together with

their category labels L_s , are computed as explained in Sec 3.1. Valid frame combinations are first sampled by selecting frame sets that satisfy the per-instance occlusion constraints. These candidate combinations are then filtered using the precomputed overlap table calculated through Eq. (5), such that C_s represents the filtered frame combinations that satisfy both occlusion and low-overlap conditions. For each task type $t \in T$, a task-specific QA generator G_t produces the corresponding set of QA pairs, Q_s . Finally, all QAs generated for the scene are merged into the global QA set.

Algorithm 2: Object Count (Single-Category)

```

1 Input: scene id  $s$ , per-scene metadata  $M_s$ ,
2 per-scene frame metadata  $F_s$ , Frame combinations  $C_s$ 
3 Output:  $Q_s$ 
4 % Select frame combination from combination sets  $C_s$ 
5 for  $C \in C_s$  do
6   % Step 1. Per-frame visible-instance extraction:
7   for  $f \in C$  do
8      $I_c(f) = \text{per\_frame\_instance}(c, f)$ 
9   end
10  % Step 2. Multi-view aggregation:
11   $V_c \leftarrow \bigcup_{f \in C} I_c(f)$ 
12   $\text{GT} \leftarrow |V_c|$ 
13  % Step 3. Multiple-choice option generation:
14   $\mathcal{D} \leftarrow \text{count\_distractor}(\text{GT})$ 
15   $\mathfrak{D} \leftarrow \{\text{GT}\} \cup \text{Sample}_3(\mathcal{D})$ 
16  % Step 4. Output assembly:
17   $Q_s = \text{filling\_qa}(\mathfrak{D}, \mathcal{T}, \text{GT})$ 
18 end

```

Object count. The Object Count task determines how many instances of an object category c are visible throughout the frame combination C . The task first identifies the per-frame visible-instances $I_c(f)$ for all frame combinations, defined as the function $I_c(f) = \{i \mid \text{cat}(i) = c, \text{occ}(i, f) \leq \tau\}$. Multi-view aggregation merges all visible instance sets across the selected frame set, forming the union V_c . The ground-truth count is then obtained as GT, representing the total number of unique instances observed. For answer choice generations, a distractor pool $\mathcal{D} = \{\max(1, \text{GT} + \delta) \mid \delta \in \{-3, -2, -1, 1, 2, 3\}\}$ generates incorrect options of a small offset from GT, where three of them are sampled as distractors in $\mathfrak{D}$. Finally, the *filling_qa* function utilizes all task-specific variables, including the question template $\mathcal{T}$, to generate the final QA instance as described in Fig. 2.

Best-view selection. The Best-view selection task determines how many instances of an object category c are visible throughout the sampled frames f , and calculates the frame f_b that gives the most informative view. The per-frame visible-instance extraction and multi-view aggregation are identical to the object count task. To select the best-view frame, we first measure how many instances of category c are visible in a frame through $n_c(f)$, then calculate the total

Algorithm 3: Object Bestview (Single-Category)

```

1 Input: scene id  $s$ , per-scene metadata  $M_s$ ,
2 per-scene frame metadata  $F_s$ , Frame combinations  $C_s$ 
3 Output:  $Q_s$ 
4 for  $C \in C_s$  do
5   % Step 1. Per-frame visible-instance extraction:
6   for  $f \in C$  do
7      $I_c(f) = \text{per\_frame\_instance}(i, c, f)$ 
8   end
9   % Step 2. Multi-view aggregation:
10   $V_c \leftarrow \bigcup_{f \in F} I_c(f)$ 
11   $\text{GT} \leftarrow |V_c|$ 
12  % Step 3. Best-view selection:
13  for  $f \in C$  do
14     $n_c(f) \leftarrow |I_c(f)|$ 
15     $A_c(f) \leftarrow \sum_{i \in I_c(f)} \text{vispix}(i, f)$ 
16  end
17   $f_b \leftarrow \arg \max_{f \in F} (n_c(f), A_c(f))$ 
18  % Step 4. Multiple-choice option generation:
19   $\mathfrak{D} = \{ (\text{GT}, f_b), (\text{GT}', f_b), (\text{GT}, f'_b), (\text{GT}', f'_b) \}$ 
20   $f'_b \leftarrow \text{random\_choice}(C \setminus \{f_b\})$ 
21  % Step 5. Output assembly:
22   $Q_s = \text{filling\_qa}(\mathfrak{D}, \mathcal{T}, \text{GT})$ 
23 end

```

visible-pixel area $A_c(f)$ for those instances computed by $\text{vispix}(i, f)$ over all visible instances i in $I_c(f)$. The best frame f_b is determined by the highest visible count; if multiple frames have the same count, ties are broken by visible-pixel comparison. Options $\mathfrak{D}$ contains one correct answer pair with the correct GT and the correct best frame f_b , while three other distractor pairs either have an incorrect count GT' from $\mathcal{D}$, an incorrect best frame f'_b from $C \setminus \{f^*\}$, or both.

Object localization. The Object Localization task determines whether a target instance i_t exists in the query frame $f_q \in C$, and returns its 2D bounding box center coordinates (x_t, y_t) if it is visible. We construct the GT by determining whether the instance is visible in the query frame using the function $\text{is_visible}(i, f) = \mathbb{1}[\text{occ}(i, f) \leq \tau]$. If the instance is visible, the GT returns a "Yes" with the instance's 2D bounding box center coordinates (x_t, y_t) , and the distractor options in $\mathfrak{D}$ consist of two positive options with incorrect coordinates $(\{(x_n, y_n)\}_{n=1}^2)$ and a negative option. If the instance is not visible, the GT returns a "No", and the distractor options in $\mathfrak{D}$ consist of three positive options with incorrect coordinates $(\{(x_n, y_n)\}_{n=1}^3)$. Incorrect coordinates are randomly sampled within the bounding box area of the target instance to avoid trivializing the task.

Algorithm 4: Object Localization (Single-Category)

```

1 Input: scene id  $s$ , per-scene metadata  $M_s$ ,
2 per-scene frame metadata  $F_s$ , Frame combinations  $C_s$ 
3 Output:  $Q_s$ 
4 for  $C \in C_s$  do
5   % Step 1. Per-frame visible-instance extraction:
6   for  $f \in C$  do
7      $I_c(f) = \text{per\_frame\_instance}(i, c, f)$ 
8   end
9   % Step 2. Multi-view aggregation:
10   $V_c \leftarrow \bigcup_{f \in F} I_c(f)$ 
11  % Step 3. Localization supervision:
12  if  $i_t$  is visible in  $f_q$  then
13     $\text{GT} \leftarrow \text{"Yes; } (x_t, y_t)\text{"}$ 
14  end
15  else
16     $\text{GT} \leftarrow \text{"No"}$ 
17  end
18  % Step 4. Multiple-choice option generation:
19  if  $\text{GT} == \text{"Yes; } (x_t, y_t)\text{"}$  then
20     $\mathfrak{D} = \{\text{GT}, \{\text{Yes; } (x_n, y_n)\}_{n=1}^2, \text{No}\}$ 
21  end
22  else  $\text{GT} == \text{"No"}$ 
23     $\mathfrak{D} = \{\text{GT}, \{\text{Yes; } (x_n, y_n)\}_{n=1}^3\}$ 
24  end
25  % Step 5. Output assembly:
26   $Q_s = \text{filling\_qa}(\mathfrak{D}, \mathcal{T}, \text{GT}, f_q)$ 
27 end

```

Multi-category tasks. Multi-category tasks share the same framework and sampling logic, where their core function is to determine the spatial relation between two object instances. After the initial per-frame visible-instance verification, the pipeline samples source and target instance pairs i_s and i_t from a randomly selected query frame f_q with the following constraints: (1) i_s and i_t cannot be the same object category, (2) the source instance i_s must be visible in the query frame f_q aligned with the predefined visibility threshold, and (3) the target instance i_t must not be visible in the query frame f_q with the same conditions. If all conditions are validated, we obtain an instance pair (i_s, i_t) for spatial inference. Directional relations R are computed by transforming a 3D oriented bounding box into the coordinate system of f_q to analyze their minimum and maximum coordinates along axis a (where $a \in \{x, y, z\}$) for strict geometric assessment.

Although all multi-category tasks share the same process up until computing the spatial relation between the instance pair (i_s, i_t) , the difference arises from constructing distractors D and options $\mathfrak{D}$:

Algorithm 5: Multi-Category Tasks

```

1 Input: scene id  $s$ , per-scene metadata  $M_s$ ,
2 per-scene frame metadata  $F_s$ , Frame combinations  $C_s$ 
3 Output:  $Q_s$ 
4 for  $C \in C_s$  do
5   % Step 1. Per-frame visible-instance extraction:
6   for  $f \in F_s$  do
7      $I_c(f) = \text{per\_frame\_instance}(i, c, f)$ 
8   end
9   % Step 2. Instance-pair construction:
10  foreach  $\text{combo}(i, C)$  do
11     $f_q \in C$ 
12     $(i_s, i_t) \leftarrow \text{select\_src\_tgt\_objects}(f_q)$ 
13  end
14  % Step 3. Spatial-relation evaluation:
15   $R \leftarrow \text{compute\_relation}(i_s, i_t, a)$ 
16  % Step 4. Question formation:
17   $D \leftarrow \text{relation\_distractor}(R)$ 
18   $\mathfrak{D} = \{R, D_1, D_2, D_3\}$ 
19  % Step 5. Output assembly:
20   $Q_s = \text{filling\_qa}(\mathfrak{D}, \mathcal{T}, \mathcal{R}, f_q, i_s, i_t)$ 
21 end

```

- **Occlusion-Aware Existence** The occlusion-aware existence task generates a question asking if the target instance exhibits a specific directional relationship from the source instance (e.g., Does the cup appear to the right of the wallet?). The corresponding options $\mathfrak{D}$ are binary, containing a "Yes" and a "No", which analyzes the model's capability of determining whether a given relation is true or not. If the relation specified in the question is equal to R from $\text{compute_relation}()$, the correct answer is "Yes"; otherwise, it is "No".
- **Occlusion-Aware Attribute** The occlusion-aware attribute task provides the spatial relation and the source instance, and asks which object satisfies the specified spatial relation with the source instance in the query frame f_q (e.g., What object appears to the left of the phone?). When constructing the options $\mathfrak{D}$, we construct an answer pool and a distractor pool: the answer pool contains the correct instance i_t , and the distractor pool contains other instances i_x in the scene that exhibit the opposite relation mentioned in the question by calculating $\text{compute_relation}(i_s, i_x, a)$. If the computed relation R is opposite to the queried relation, i_x is added to the distractor pool. In $\mathfrak{D}$, we sample one option from the answer pool (containing the target instance) and the remaining three from the distractor pool, ensuring that only one object satisfies the query condition. Note that constructing valid multiple-choice options for the attribute task requires at least four distinct object categories to be visible within the given frames. Since WOD contains only

four annotated object categories in total, we do not generate attribute-based VQA samples for the outdoor subset.

- **Occlusion-Aware Spatial Relation** The occlusion-aware spatial relation task provides the source instance, target instance, and query frame f_q , and asks which spatial relation holds between the two objects (e.g., Where does the pillow appear relative to the door?). Upon evaluation on a specific axis a , we first calculate the true spatial relation R between i_s and i_t using *compute_relation()*. The resulting relation R serves as the correct answer and is stored in the answer pool. The distractor pool consists of spatial relations that differ from the true relation between i_s and i_t . Therefore, the first distractor D_1 is the opposite relation of R along the same axis a . The other two distractors D_2 and D_3 are derived by computing the relations between i_s and i_t along the two remaining orthogonal axes b and c . Since R_b and R_c represent the true relations along axes b and c , their opposite relations are added to the distractor pool. For example, if the target instance is located "to the left", "below", and "farther" from the source instance, and the evaluation is performed along the x-axis (left/right), the answer pool contains "left", while the distractor pool contains "right", "above", and "closer". However, this construction may introduce a potential axis-based bias, since when two relations from the same axis are included, one of them must be correct. To mitigate this issue, we additionally construct a bias-reduced VQA variant, where distractors are not sampled from the same axis as the correct relation. As illustrated in Fig. 15 and 20, this variant includes only relations from orthogonal axes, resulting in three-option multiple-choice questions that prevent axis-based shortcuts.

F Architecture Details of SpatialMosaicVLM

SpatialMosaicVLM constructs a joint representation of a 3D scene by extracting visual tokens and geometric tokens. These tokens are subsequently fused via cross attention to form a unified representation, which serves as input to the language model backbone. Given multi-view images $\{\mathbf{I}_v\}_{v=1}^V$, we employ two encoders: a visual encoder E_{vis} and a geometric encoder E_{geo} .

Visual encoder. We use a pretrained CLIP ViT as the visual encoder. For each view $\mathbf{I}_v$, the vision encoder produces a sequence of patch-level visual tokens $F_{vis}^{(v)}$:

$$F_{vis}^{(v)} = E_{vis}(\mathbf{I}_v) \in \mathbb{R}^{T_{vis}^{(v)} \times d}, \quad (8)$$

where $T_{vis}^{(v)}$ is the number of visual tokens per view and d is the feature dimension. We aggregate the visual tokens across all views by concatenating them along the token dimension to obtain the global visual token set:

$$F_{vis} = [F_{vis}^{(1)}; F_{vis}^{(2)}; \dots; F_{vis}^{(V)}] \in \mathbb{R}^{T_{vis} \times d}. \quad (9)$$

Geometric encoder. For geometric encoding, we adopt VGGT as E_{geo} , which jointly processes multi-view images to recover scene-level geometric structure.

Given the multi-view images, the encoder yields spatial features and camera tokens:

$$(F_{spa}, z) = E_{geo}(\{\mathbf{I}_v\}_{v=1}^V), \quad (10)$$

where $F_{spa} \in \mathbb{R}^{T_{spa} \times d}$ denotes spatial features, and $z \in \mathbb{R}^{V \times d}$ denotes camera tokens. Then, We concatenate spatial features F_{spa} and camera tokens z to obtain the geometric tokens F_{geo} :

$$F_{geo} = [F_{spa}; z] \in \mathbb{R}^{(T_{spa}+V) \times d}. \quad (11)$$

Finally, the visual tokens F_{vis} and geometric tokens F_{geo} are fused through cross attention as described in Sec. 4. The fused representation is then concatenated with the language tokens and fed into the language model backbone for answer generation.

G Experimental setting

As described in Sec. 3, we construct the SpatialMosaic dataset using ScanNet++ [38] and WOD [29], spanning both indoor and outdoor scenes. We select 757 scenes for training and 179 for testing. To enable rich and diverse VQA generation, we utilize all annotated object categories present in each scene. Moreover, to ensure that the generated QA pairs cover a wide variety of target-source object combinations, each target object category is explicitly paired with multiple source-object categories. This pairing strategy enables the pipeline to generate a large and diverse set of QA pairs across a wide variety of object pairs. For a fair comparison, all ablated models are fine-tuned under identical training settings using the training splits of SpatialMosaic and VSI-Bench, where the latter follows the data generation procedure described by VSI-Bench. For training, it requires 8 NVIDIA H200 GPUs with a batch size of 4 for one epoch, which takes approximately 16 hours per model. To conduct an evaluation on our *SpatialMosaic-Bench*, we compare with various open-source MLLM baselines, with inference time ranging from 1 to 5 hours depending on model size. Since full-scale comparison experiments are extremely time-intensive even on 8 NVIDIA H200 GPUs, we conduct the fine-tuning and quantitative evaluation in the main paper using a reduced subset of the dataset consisting of 200K training samples and 100K test samples.

H SpatialMosaic Data Samples

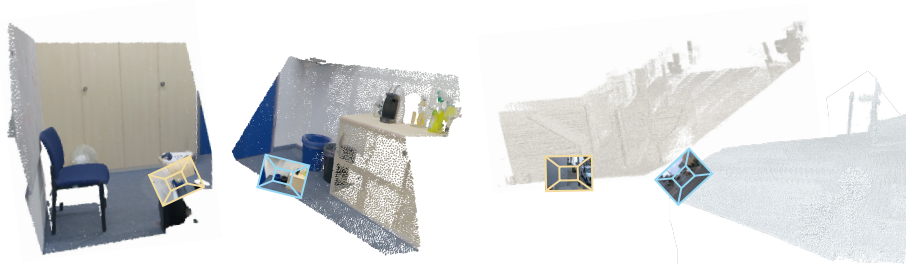
SpatialMosaic dataset provides 12 sub-tasks. We provide the QA examples from both indoor and outdoor scenes, as shown in Fig. 10 - 20.

I SpatialMosaic VQA Templates

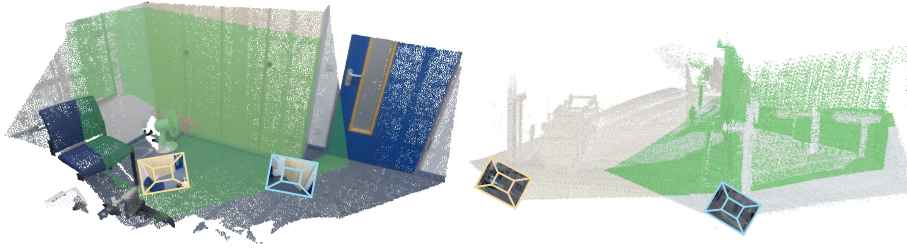
We leverage an automated VQA generation pipeline to construct extensive question-answer pairs. The corresponding templates for each task are listed in Table 7.

Table 7: Question templates used in the **SpatialMosaic** benchmark. LR: Left/Right, AB: Higher/Lower, FB: Closer/Farther.

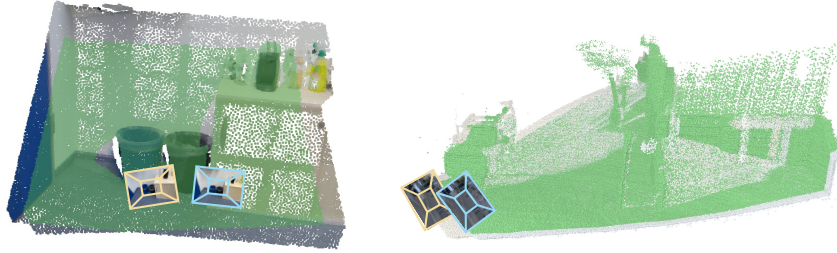
Task	Question Template	Answer Type
Object Count	How many {category}(s) are visible across these frames?	{Number}
Best View Selection	How many {category}(s) are visible across these frames? And tell me which frame provides the most informative view of the {category}(s)?	{Number} + {Frame ID}
Existence (LR)	In {frame_id}, does the {object1} appear {to the left of / to the right of} the {object2} in this viewpoint?	{Yes/No}
Existence (AB)	In {frame_id}, does the {object1} appear {higher than / lower than} the {object2} in this viewpoint?	{Yes/No}
Existence (FB)	In {frame_id}, does the {object1} appear {closer to / farther from} the camera than the {object2} in this viewpoint?	{Yes/No}
Attribute (LR)	In {frame_id}, which object appears {to the left of / to the right of} the {object} in this viewpoint?	{Object name}
Attribute (AB)	In {frame_id}, which object appears {higher than / lower than} the {object} in this viewpoint?	{Object name}
Attribute (FB)	In {frame_id}, which object appears {closer to / farther from} the camera than the {object} in this viewpoint?	{Object name}
Spatial Relation (LR / AB / FB)	In {frame_id}, where does the {target} appear in this view relative to the {source}?	{Spatial Rel.}
Localization	Is there a(n) {target} in {frame_id}? If so, what is the bounding box center coordinates?	{Coordinates}



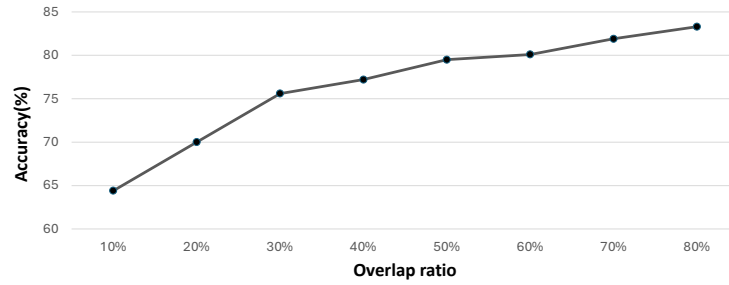
(a) overlap ratio: 0% (left: indoor, right: outdoor)

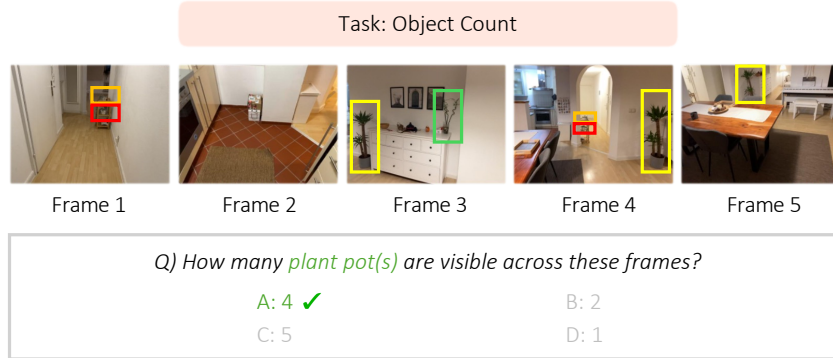
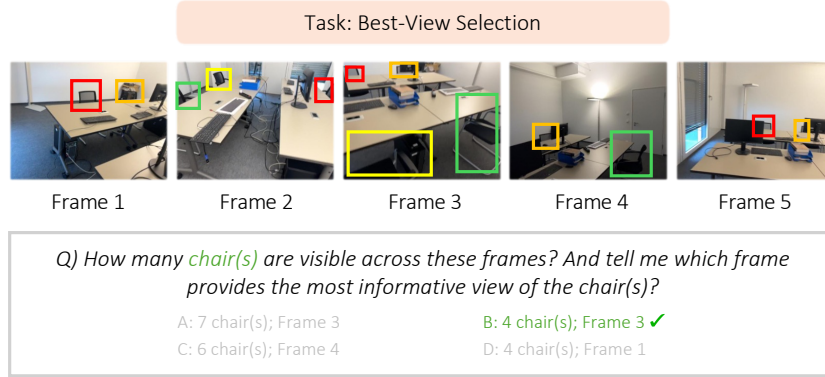
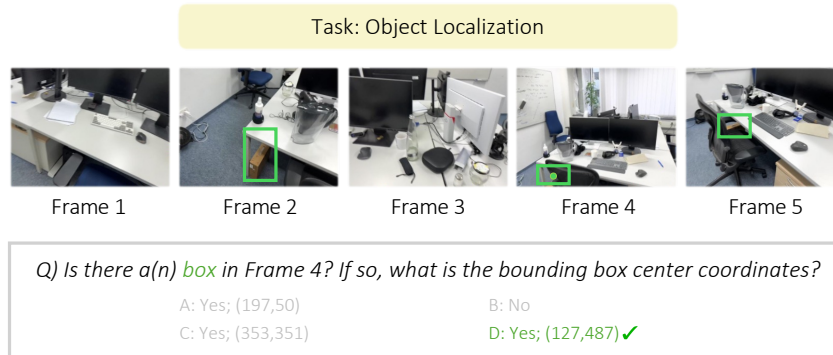


(b) overlap ratio: 30% (left: indoor, right: outdoor)



(c) overlap ratio: 60% (left: indoor, right: outdoor)

Fig. 8: Qualitative examples of varying overlap ratios**Fig. 9:** Quantitative analysis on different overlap ratios

**Fig. 10:** Object Count example**Fig. 11:** Best-View Selection example**Fig. 12:** Object Localization example

Task: Occlusion-Aware Object Existence (left / right)



Frame 1



Frame 2



Frame 3



Frame 4

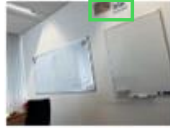
Q) In Frame 1, does the *printer* appear to the right of the *plant* in this viewpoint?

A: Yes ✓

B: No

(a) left / right

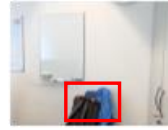
Task: Occlusion-Aware Object Existence (higher / lower)



Frame 1



Frame 2



Frame 3



Frame 4

Q) In Frame 3, does the *picture* appear higher than the *jacket* in this viewpoint?

A: No

B: Yes ✓

(b) higher / lower

Task: Occlusion-Aware Object Existence (farther / closer)



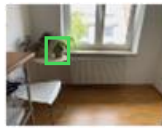
Frame 1



Frame 2



Frame 3



Frame 4

Q) In Frame 2, does the *plant* appear farther from the camera than the *container* in this viewpoint?

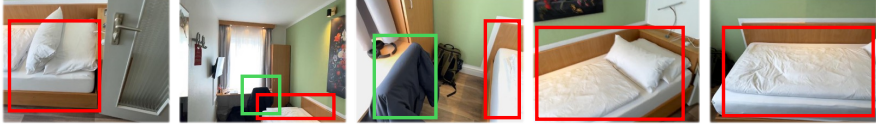
A: No

B: Yes ✓

(c) farther / closer

Fig. 13: Occlusion-Aware Object Existence examples

Task: Occlusion-Aware Attribute (left / right)



Frame 1 Frame 2 Frame 3 Frame 4 Frame 5

Q) In Frame 1, which object appears to the left of the *bed* in this viewpoint?

A: Shower floor B: Jacket ✓
C: Door D: Pillow

(a) left / right

Task: Occlusion-Aware Attribute (higher / lower)



Frame 1 Frame 2 Frame 3 Frame 4 Frame 5

Q) In Frame 5, which object appears higher than the *headphone case* in this viewpoint?

A: Chair B: Heater
C: White board ✓ D: Table

(b) higher / lower

Task: Occlusion-Aware Attribute (farther / closer)



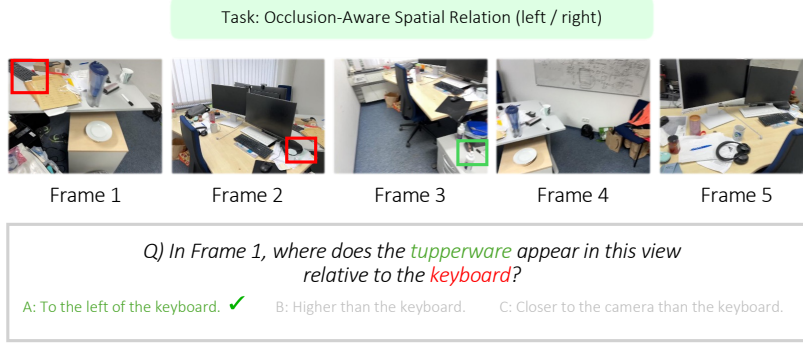
Frame 1 Frame 2 Frame 3 Frame 4

Q) In Frame 1, which object appears closer to the camera than the *telephone* in this viewpoint?

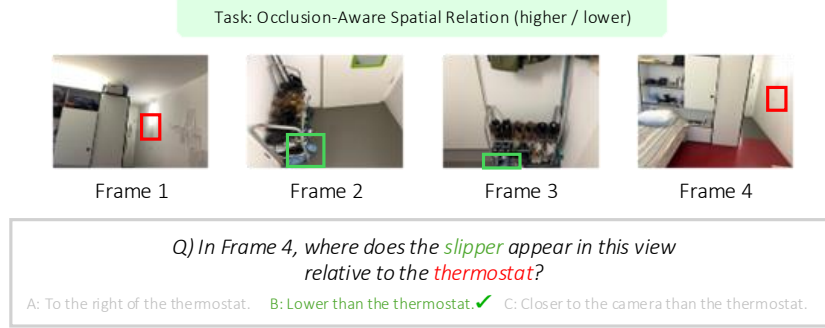
A: A trash bin B: Window
C: Power socket D: Bookshelf ✓

(c) farther / closer

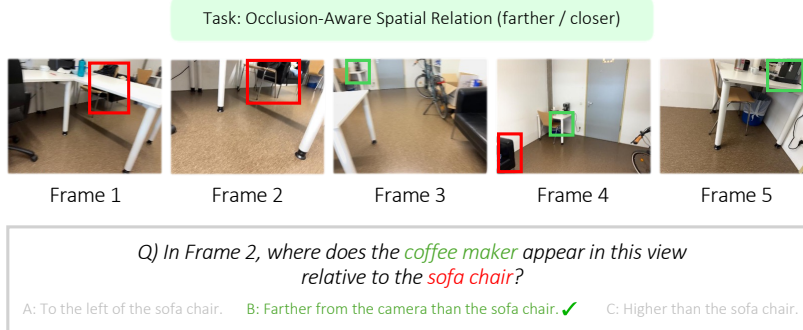
Fig. 14: Occlusion-Aware Attribute examples



(a) left / right



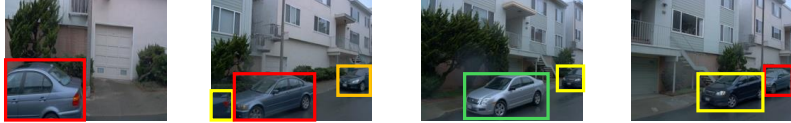
(b) higher / lower



(c) farther / closer

Fig. 15: Occlusion-Aware Spatial Relation examples

Task: Object Count



Frame 1 Frame 2 Frame 3 Frame 4

Q) How many *vehicle(s)* are visible across these frames?

A: 4 ✓ B: 2
C: 5 D: 6

Fig. 16: Object Count example (outdoor)

Task: Best-View Selection



Frame 1 Frame 2 Frame 3 Frame 4

Q) How many *vehicle(s)* are visible across these frames? And tell me which frame provides the most informative view of the vehicle(s).

A: 7 vehicle(s); Frame 3 B: 6 vehicle(s); Frame 3 ✓
C: 6 vehicle(s); Frame 4 D: 4 vehicle(s); Frame 1

Fig. 17: Best-View Selection example (outdoor)

Task: Object Localization



Frame 1 Frame 2 Frame 3 Frame 4

Q) Is there a(n) *pedestrian* in Frame 4?
If so, what is the bounding box center coordinates?

A: Yes; (488,277) B: No
C: Yes; (504,301) D: Yes; (492,298) ✓

Fig. 18: Object Localization example (outdoor)

Task: Occlusion-Aware Object Existence (left / right)



Q) In Frame 1, does the *vehicle* appear to the right of the *sign(370,223)* in this viewpoint?

A: Yes ✓

B: No

(a) left / right

Task: Occlusion-Aware Object Existence (higher / lower)



Q) In Frame 1, does the *pedestrian* appear higher than the *vehicle* in this viewpoint?

A: No

B: Yes ✓

(b) higher / lower

Task: Occlusion-Aware Object Existence (farther / closer)



Q) In Frame 2, does the *pedestrian* appear closer to the camera than the *vehicle(383,445)* in this viewpoint?

A: Yes

B: No ✓

(c) farther / closer

Fig. 19: Occlusion-Aware Object Existence examples (outdoor)

Task: Occlusion-Aware Spatial Relation (left / right)



Frame 1 Frame 2 Frame 3 Frame 4

Q) In Frame 1, where does the *sign* appear in this view relative to the *vehicle*(67,345)?

A: Lower than the vehicle. B: To the right of the vehicle. ✓ C: Farther from the camera than the vehicle.

(a) left / right

Task: Occlusion-Aware Spatial Relation (higher / lower)



Frame 1 Frame 2 Frame 3 Frame 4

Q) In Frame 1, where does the *pedestrian* appear in this view relative to the *sign*?

A: Lower than the sign. ✓ B: To the right of the sign. C: Closer to the camera than the sign.

(b) higher / lower

Task: Occlusion-Aware Spatial Relation (farther / closer)



Frame 1 Frame 2 Frame 3 Frame 4

Q) In Frame 1, where does the *vehicle* appear in this view relative to the *sign*(354,220)?

A: Lower than the sign. B: Closer to the camera than the sign. ✓ C: To the right of the sign.

(c) farther / closer

Fig. 20: Occlusion-Aware Spatial Relation examples (outdoor)